

Irfan Ahmad Ganie

Assistant Professor, Mechanical & Electrical Engineering

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EDUCATION

Ph.D. Electrical Engineering

2025

Missouri University of Science and Technology (MS&T), Rolla, MO, USA

Thesis title: Human-Robot Teaming using Safe Lifelong Learning based

Optimal Adaptive Control Framework

Advisor: Sarangapani Jagannathan

2021

MTech. Cyber-Physical System

Indian Institute of Technology (IIT), Jodhpur, Rajasthan, India

Thesis title: Integral Sliding Mode Control for Non-Linear Systems

Advisor: Deepak Fulwani

2018

B.E. Electrical Engineering

University of Jammu, India

PROFESSIONAL APPOINTMENTS

Assistant Professor

Aug 2025-Present

Mechanical & Electrical Engineering, Wilkes University, Wilkes-Barre, PA, USA

Research Associate

May-Aug 2025

Department of Electrical and Computer Engineering,

Missouri University of Science & Technology, Rolla, MO, USA

Supported Projects funded by Army Research Office (ARO), Office of Naval Research (ONR), AND Air Force Office of Scientific Research (AFOSR)

Course Instructor

Jan-May 2025

Control Systems,

Missouri University of Science and Technology, Rolla, MO, USA

Graduate Research Assistant

2021-2025

Department of Electrical and Computer Engineering,

Missouri University of Science and Technology, Rolla, MO, USA

Supported Projects funded by Army Research Office (ARO), Office of Naval Research (ONR), AND Air Force Office of Scientific Research (AFOSR)

HONORS AND AWARDS

- **2025: Deans Scholar Award**, College of Engineering, Missouri S&T (2025)

One of six Ph.D. students recognized for outstanding scholarly contributions and exceptional research productivity within the College of Engineering.

- **2023: Intelligent Systems Center (ISC) Poster Presentation Award Winner,** *Missouri University of Science and Technology, College of Engineering and Computing.*
- **2023: IEEE Control Systems Society Conference Travel Award,** *Institute of Electrical and Electronics Engineers (IEEE)*

TEACHING

Undergraduate and Graduate Courses, Wilkes University, PA, USA

2025-present

EE-314 Control Systems (S'26)
 EE314 B1 Control Systems LAB (S'26)
 EGR-214 Linear Systems (S'26)
 EGR-214 A Linear Systems LAB (S' 26)
 EE-498 Artificial Intelligence-based Control for Robotic Systems (SP'26).
 EE-241: Digital Design (SP' 26),
 EE-241-&HA: Digital Design (Honors) (SP'26)
 EE-241-L1 / EE-241-&HL: Digital Design Laboratory (SP'26)
 EE-252: Electronics II (F'25)
 EE-252-L1: Electronics II Laboratory (F'25)
 EE-283: Electrical Engineering Laboratory (F'25)

Graduate Courses, Missouri S&T, Rolla, MO, USA

Jan-May 2025

Mech 6410/Aero 6410/ EE 6310: Optimal Control and Estimation (S' 25)

RESEARCH INTERESTS

- Online Learning–Based Intent Estimation and Optimal Cooperative Control for Human–Robot Teaming and Physical Interaction
- Online Neuromorphic Deep Learning Control Framework for Nonlinear Systems.
- Trustworthy and Explainable AI–Based Adaptive Controllers for Cyber–Physical Learning Systems.
- Adaptive Continual Learning with Safety Awareness for Nonlinear Systems

GRANT PROPOSALS

- **PI (NSF DCSD Proposal, Pending):** RUI: Lyapunov-Based Deep Graph Neural Networks for Safe Cooperative Optimal Control of Non-Stationary Interconnected Systems— \$149K, Wilkes University (2027–2029).
- **PI (NSF CPS Proposal, Pending):** Energy-Efficient Safety-Critical Adaptive Optimal Control using Spiking Neural Networks — \$225K, Wilkes University (2027–2029).
- **PI (Funded)** Summer Mentoring Grant, Wilkes University-Student Research Mentorship— \$9559 (2026)

PUBLICATIONS [Google Scholar](#)

Articles Under Review

13. **I. Ganie**, S. Jagannathan “Online Learning Driven Human Intent Estimation and Game Theoretic Control for Human Robot Interaction with Safety Guarantees” 2026, under review *in*

Peer-reviewed Journal Publications

12. **I. Ganie**, S. Jagannathan. “. Safety Aware Deep Neural-based Distributed Optimal Control for Cooperative Manipulation of Human Robot Teams” 2025, in *IEEE Transactions on Cybernetics*, 2026. [doi: 10.1109/TCYB.2026.3651182](https://doi.org/10.1109/TCYB.2026.3651182),
11. **I. Ganie**, S. Jagannathan. “Safety Aware Continual Reinforcement Learning-based Output Tracking Control of Nonlinear Continuous-time Systems” in *IEEE Transactions on System's man and Cybernetics*, 2026. [doi: 10.1109/TSMC.2025.3647584](https://doi.org/10.1109/TSMC.2025.3647584)
10. **I. Ganie**, and S. Jagannathan, “Online Lifelong Optimal Tracking Control of Nonlinear Continuous-Time Strict-Feedback Systems using Deep Neural Networks,” *Neural Networks*, vol. 191, 107793, 2025, ISSN 0893-6080. <https://doi.org/10.1016/j.neunet.2025.107793>
9. E. Soleimani, **I. Ganie**, and S. Jagannathan, “Safe Optimal Control of Quadrotor Formations Using Multilayer Neural Networks and Continual Learning,” *International Journal of Adaptive Control and Signal Processing*, vol. 39, no. 8, pp. 1744–1766, 2025. <https://doi.org/10.1002/acs.4020>
8. **I. Ganie**, and S. Jagannathan, “Lifelong Learning-Based Optimal Trajectory Tracking of Constrained Nonlinear Affine Systems using Deep Neural Networks,” *IEEE Transactions on Cybernetics*, pp. 1–14, 2024. <https://doi.org/10.1109/TCYB.2024.3405354>
7. **I. Ganie**, and S. Jagannathan, “Optimal Trajectory Tracking of Uncertain Nonlinear Continuous-Time Strict-Feedback Systems with Dynamic Constraints,” *International Journal of Control*, pp. 1–15, 2024. <https://doi.org/10.1080/00207179.2024.2370323>
6. **I. Ganie**, and S. Jagannathan, “Lifelong Reinforcement Learning Tracking Control of Nonlinear Strict-Feedback Systems using Multilayer Neural Networks with Constraints,” *Neurocomputing*, vol. 600, pp. 128–139, 2024. <https://doi.org/10.1016/j.neucom.2024.128139>
5. **I. Ganie**, and S. Jagannathan, “Continual Online Learning-Based Optimal Tracking Control of Nonlinear Strict-Feedback Systems: Application to Unmanned Aerial Vehicles,” *Complex Engineering Systems*, vol. 4, no. 1, 2024. <https://doi.org/10.20517/ces.2023.35>
3. **I. Ganie**, and S. Jagannathan, “Lifelong Deep Learning–Based Control of Robot Manipulators,” *International Journal of Adaptive Control and Signal Processing*, vol. 37, no. 12, pp. 3169–3192, 2023. <https://doi.org/10.1002/acs.3678>
2. **I. Ganie**, and S. Jagannathan, “Lifelong Learning-Based Multilayer Neural Network Control of Nonlinear Continuous-Time Strict-Feedback Systems,” *International Journal of Robust and Nonlinear Control*, vol. 34, no. 2, pp. 1397–1416, 2023. <https://doi.org/10.1002/rnc.7039>
2. M. Shabir, J. Din, and **I. Ganie**, “Multi-Granulation Roughness Based on Soft Relations,” *Journal of Intelligent and Fuzzy Systems*, vol. 40, no. 6, pp. 10893–10908, 2021. <https://doi.org/10.3233/JIFS-201910>
1. A. E. Kurika, **I. Ganie**, Y. Kadir, P. D. Cerna, and F. L. Desei, “Predicting Factors of Vehicular Accidents Using Machine Learning Algorithm,” *International Journal of Emerging Trends in Engineering Research*, vol. 8, no. 9, pp. 5171–5176, September 2020. <https://doi.org/10.30534/ijeter/2020/46892020>

Conference Publications

12. **I. Ganie**, S. Jagannathan “Energy-Efficient Intent Estimation and Distributed Optimal Control for Human–Multi-Robot Collaboration” in American Control Conference (ACC) 2026 (Accepted).

11. **I. Ganie**, S. Jagannathan “Safety-Critical Adaptive Spiking Multilayer Neural Network Control of Nonlinear Systems” in American Control Conference (ACC) 2026 Accepted for publication.
- 10 **I. Ganie** and S. Jagannathan, “Optimal Control for Human–Robotic Agent Teams for Cooperative Manipulation,” *Proceedings of the 2025 IEEE Conference on Decision and Control (CDC)*, Rio, Brazil, December 2025.
- 9 M. Geiger, **I. Ganie**, and S. Jagannathan, “Finite-Horizon Optimal Adaptive Tracking of Uncertain Linear Discrete-Time Systems Using Q-Learning,” *Proceedings of the IFAC Joint Conference on Computers, Cognition and Communication (J3C 2025)*, Padova, Italy, 2025.
- 8 **I. Ganie** and S. Jagannathan, “Optimal Continual Reinforcement Learning–Based Optimal Output Tracking Control of Nonlinear Systems Using Multilayer Observer,” *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025, pp. 1-6, doi: 10.1109/IJCNN64981.2025.11229184.
- 7 **I. Ganie** and S. Jagannathan, “Online Learning-Driven Human Intent Estimation and Control for Human–Robot Interaction,” *Proceedings of the 2025 American Control Conference (ACC)*, Denver, CO, USA, 2025, pp. 5160–5165, doi: 10.23919/ACC63710.2025.11107829.
- 6 **I. Ganie** and S. Jagannathan, “Online Continual Safe Reinforcement Learning–Based Optimal Control of Mobile Robot Formations,” *Proceedings of the 2024 IEEE Conference on Control Technology and Applications (CCTA)*, Newcastle upon Tyne, United Kingdom, 2024, pp. 519–524, doi: 10.1109/CCTA60707.2024.10666606.
- 5 **I. Ganie** and S. Jagannathan, “Continual Optimal Adaptive Tracking of Uncertain Nonlinear Continuous-Time Systems Using Multilayer Neural Networks,” *Proceedings of the 2023 American Control Conference (ACC)*, San Diego, CA, USA, 2023, pp. 3395–3400, doi: 10.23919/ACC55779.2023.10156466.
- 4 **I. Ganie** and S. Jagannathan, “Lifelong Learning Control of Nonlinear Systems with Constraints Using Multilayer Neural Networks with Application to Mobile Robot Tracking,” *Proceedings of the 2023 IEEE Conference on Control Technology and Applications (CCTA)*, Bridgetown, Barbados, 2023, pp. 727–732, doi: 10.1109/CCTA54093.2023.10252103.
- 3 **I. Ganie** and S. Jagannathan, “Adaptive Control of Robotic Manipulators Using Deep Neural Networks,” *Proceedings of the 6th IFAC Conference on Intelligent Control and Automation Sciences (ICONS 2022)*, vol. 55, no. 15, pp. 148–153, 2022, doi: 10.1016/j.ifacol.2022.07.623
- 2 **Irfan Ahmad** and K. Pothuganti, “Smart Field Monitoring Using ToxTrac: A Cyber-Physical System Approach in Agriculture,” *IEEE International Conference on Smart Electronics and Communication (ICOSEC)*, 2020.
- 1 **Irfan Ahmad** and K. Pothuganti, “Design and implementation of real-time autonomous car using image processing and IoT,” *Proc. 2020 Third Int. Conf. Smart Systems and Inventive Technology (ICSSIT)*, pp. 107–113, Aug. 2020.

PRESENTATIONS

Keynotes/Invited Talk

“From Commands to Cooperation: Designing Intelligent Human–Robot Teams with AI”
Invited Speaker, 7th Annual tecBRIDGE Innovation Conference, Scranton, PA, 2025.

Poster Presentations

- 2 “Continual Learning–Based Formation Control,” presented at the Intelligent Systems Center Poster Presentation, Missouri University of Science and Technology, Rolla, MO, 2022.
1. “Online Deep Reinforcement Learning–Based Tracking Control for Leader–Follower Mobile Robot Formation,” presented at the Intelligent Systems Center Poster Presentation, Missouri University of Science and Technology, Rolla, MO, 2023.

STUDENT THESIS ADVISOR/COMMITTEE.

- Committee Member – M.S. Mechanical Engineering (Nick Romas), Wilkes University, 2026

PROFESSIONAL SERVICES.

University Committees

- Student Advisor, *in Electrical Engineering, Wilkes University* 2025–Present
- Chair, Research Collaboration and Symposium Committee, *College of Business and Engineering, Wilkes University* 2025–Present
- Library Committee, *Wilkes University* 2026–Present

Reviewing Service 2022–Present

- Served as a reviewer for leading IEEE journals and conferences in control systems, automation, and intelligent learning systems, including:
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Systems, Man, and Cybernetics
- IEEE Transactions on Control System Technology.
- IEEE Transactions on Automation Science and Engineering
- IEEE Control Systems Society Conferences:
- *American Control Conference (ACC)*
- IEEE Conference on Control Technology and Applications